# 3.5 Equations of Parallel and Perpendicular Lines

Name _____ Period ____ Date _____

Geometry 3.5 · Student sheet

**I can…**

- find a point that partitions a segment in a given ratio.
- use slopes to tell whether lines are parallel or perpendicular.
- write equations of parallel and perpendicular lines.

**Vocabulary** · fill in the missing word

**Slope** — the ratio of the _____ change (rise) to the horizontal change (run) between two points on a line

**Directed line segment** — a segment that has a direction from one _____ to the other

## Part 1 · Discovery: slopes that match

Graph $y = 2x + 1$ and $y = 2x - 3$. Use a square corner to draw a perpendicular line through $(0, 2)$. Find its slope.

| Line | Slope | Slope × 2 |
|---|---|---|
| $y = 2x + 1$ | | |
| $y = 2x - 3$ | | |
| Perpendicular line | | |

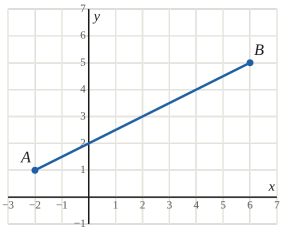
The slopes of perpendicular lines multiply to _____.

## Part 2 · Notes and examples (fill in each note, then work the example)

**Partitioning a segment:** the point k of the way from $A(x_1, y_1)$ to $B(x_2, y_2)$ is $(x_1 + k(x_2 - x_1),$ _____$)$.

### Example 1

Find the point P that is $\frac{3}{4}$ of the way from $A(-2, 1)$ to $B(6, 5)$.




**Slopes of Parallel Lines Theorem:** two nonvertical lines are parallel if and only if they have the _____ slope.

### Example 2

Write an equation of the line through $(3, 4)$ that is parallel to $y = 2x - 1$.

**Slopes of Perpendicular Lines Theorem:** two nonvertical lines are perpendicular if and only if the product of their slopes is _____.

### Example 3

Write an equation of the line through (2, 5) that is perpendicular to $y = 2x - 1$.

### Example 4

Line a passes through (0, 1) and (2, 5). Line b passes through (1, −1) and (5, −3). Are they parallel, perpendicular, or neither?

**Distance from a point to a line:** write the perpendicular line through the point, find the _____ of the two lines, then use the Distance Formula.

### Example 5

Find the distance from the point (5, 0) to the line $y = x + 1$. Round to the nearest hundredth.

## Part 3 · Partner practice

### Partner 1

Write an equation of the line through (1, 2) parallel to $y = -3x + 4$.

### Partner 2

Write an equation of the line through (4, −1) perpendicular to $y = (\frac{2}{3})x + 1$.